

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B083-Networkwalks
Date	17 SEPTEMBER 2026
Modules completed	W2-PM2 (GHDB Footprinting) W2-PM4 (theHarvester Footprinting) W2-PM5 (Zenmap Network Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes

1. LIABILITY DISCLAIMER

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2. INTRODUCTION

This report covers footprinting and network scanning activities completed during Week 2 of the Cybersecurity & Ethical Hacking internship program at Networkwalks. The activities demonstrate the reconnaissance phase of a penetration test, including:

- GHDB (Google Hacking Database) footprinting to identify exposed devices and resources
- theHarvester reconnaissance against target domains to collect email addresses and subdomains
- Zenmap network scanning to identify and map live hosts on a local network

Each module covers a different aspect of information gathering - from passive reconnaissance using public sources to active network discovery. Every step includes the exact commands used, results observed, screenshots as evidence, and analysis of security implications.

3. TOOLS USED

Tool	Purpose
Google	Search engine for GHDB dork exploitation
GHDB	Pre-made Google search queries for footprinting
theHarvester	Email and subdomain collection tool (Python-based)
Kali Linux	Operating system for reconnaissance tools
Zenmap	GUI version of Nmap for network scanning

4. W2-PM2: FOOTPRINTING WITH GHDB

4.1 Background

GHDB (Google Hacking Database) is a collection of ready-made search queries called 'dorks' that use normal Google search operators in clever ways to pull out sensitive information that websites have accidentally left public. These include exposed camera feeds, open directories, login pages, config files, and documents. GHDB is hosted on www.exploit-db.com and was built for pentesters and security researchers to find and fix these leaks.

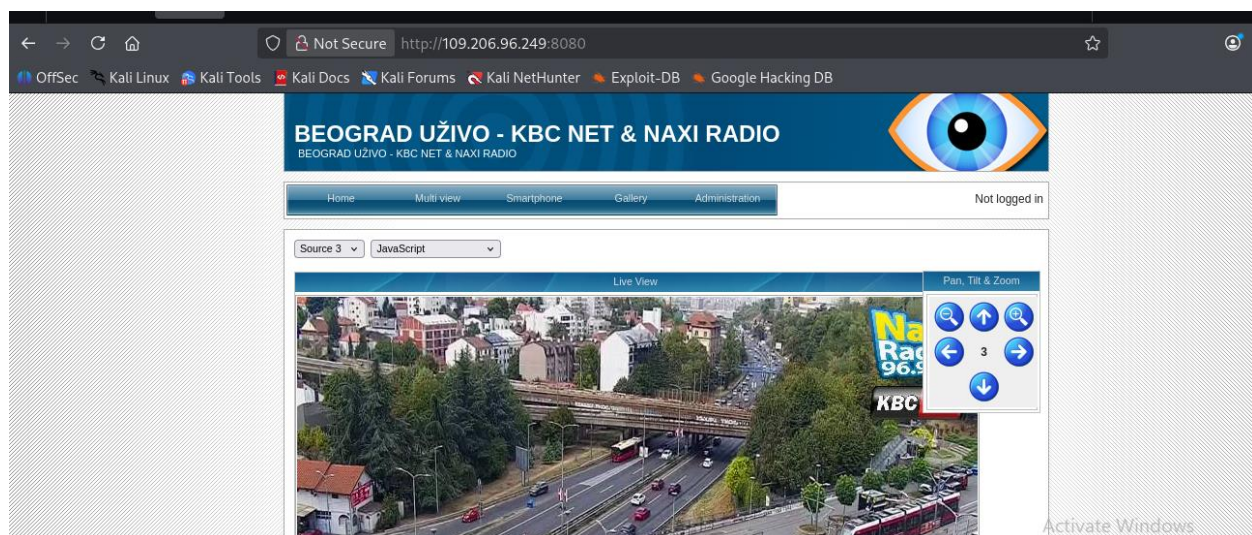
4.2 Task 1: Exposed Security Cameras

Objective: Find 10 live vulnerable security camera links exposed on the internet using GHDB dorks.

No.	Camera Link	Relevant Dork
1	http://99.114.240.169	intitle:"Webcam" inurl:WebCam.htm
2	https://copyrighthistory.org/cam/tools	intitle:"Index of /cam/"
3	https://boulderjct.org/webcam/?C=N;O=D	intitle:"Index of /webcam/"
4	https://ns.ph.liv.ac.uk/webcam/	intitle:"Index of /webcam/"
5	http://mediaplace.ath.forthnet.gr:81/	intitle:"Device(IP CAMERA)" "language"

Etc.....

2023-11-07	intitle:webcam inurl:webcam.inurl	Various Online Devices	s Thakur
2023-03-14	Google Dork : intitle:"Toshiba Network Camera"	Various Online Devices	Shilpa Shaji
2023-02-27	intitle:"Index of /cam/"	Files Containing Juicy Info	Sanu Jose M
2023-01-31	intitle:"Index of /webcam/"	Files Containing Juicy Info	Shuvrosayar Das
2022-07-25	intitle:"Device(IP CAMERA)" "language" -com.net	Pages Containing Login Portals	s Thakur
2022-07-25	intitle:"NoVus IP camera" -com	Various Online Devices	s Thakur
2022-07-20	intitle:"Network Camera" inurl:main.cgi	Various Online Devices	s Thakur
2022-06-17	inurl:webcam site:skylinewebcams.com inurl:roma	Various Online Devices	Simone Gasparato
2021-11-16	intitle:"Login" intext:"cam"	Pages Containing Login Portals	Yash Singh
2021-11-16	intext:"Real-time IP Camera Monitoring System" intext:"ActiveX Mode (For IE Browser)"	Various Online Devices	Yash Singh
2021-11-12	intitle:"Login" intext:"camera"	Pages Containing Login Portals	Naveen Venugopal
2021-11-09	intitle:"webcamXP" inurl:8080	Various Online Devices	Krishna Agarwal
2021-11-09	intitle:"Index of" "DCIM/camera"	Files Containing Juicy Info	Krishna Agarwal
2021-10-28	intitle:ip camera login page	Pages Containing Login Portals	jawhar milan
2021-10-26	intitle:"Microseven M7CAM IP Camera"	Various Online Devices	activate Windows Neha Singh



4.3 Task 2: Mathematics eBooks

Objective: Find 10 listings containing downloadable mathematics eBooks in PDF format.

No.	eBook Location Link	Relevant Dork
-----	---------------------	---------------

1	https://www.skylineuniversity.ac.ae/pdf/math/	intitle:index.of "parent directory" mathematics pdf
2	http://erewhon.superkuh.com/library/Math/	intitle:index.of "parent directory" mathematics pdf
3	https://www.unm.edu/~megrad/Math/	intitle:index.of "parent directory" mathematics pdf
4	https://berkeley.edu	intitle:"index of" ext:pdf "advanced mathematics"
5	https://www.aerostudents.com/courses/calculus-1-b/	intitle:"index of /" +mathematics +calculus pdf

Etc.....

Index of /~megrad/Math

<u>Name</u>	<u>Last modified</u>	<u>Size</u>	<u>Description</u>
Parent Directory		-	
Fall_2007.pdf	29-Jan-2009 15:31	45K	
Fall_2008.doc	29-Jan-2009 16:55	64K	
Fall_2009.pdf	15-Sep-2009 08:28	39K	
Fall_2010.pdf	22-Sep-2010 18:12	111K	
Fall_2011.pdf	23-Sep-2011 05:59	116K	
Mathematics-2017.pdf	19-Feb-2018 12:28	79K	
Spring_2005.pdf	29-Jan-2009 16:57	30K	
Spring_2008.pdf	29-Jan-2009 16:50	28K	
Spring_2009.pdf	27-Mar-2009 12:46	61K	
Spring_2010.pdf	06-Apr-2010 09:34	89K	
Spring_2011.pdf	01-Jun-2011 08:13	142K	
Spring_2012.pdf	16-Mar-2012 08:45	232K	
Spring_2013.pdf	08-May-2013 07:06	86K	

4.4 Findings & Impact

20 sensitive resources were identified through passive reconnaissance using Google alone. These findings demonstrate that:

- Organizations unknowingly expose security cameras, often with no authentication
- Educational institutions leave open directory listings containing valuable intellectual property
- Search engines index more sensitive content than most organizations realize

5. W2-PM4: FOOTPRINTING WITH theHARVESTER

5.1 Background

theHarvester is a Python-based footprinting tool that gathers information about target organizations from public sources including search engines, PGP key servers, and SHODAN databases. It collects emails, subdomains, hosts, employee names, open ports and banners without directly contacting the target.

5.2 Task 1: Reconnaissance with Baidu Source

Command Executed:

```
$ theHarvester -d microsoft.com -l 1000 -b baidu
```

Results:

- Target: microsoft.com
- Source: Baidu search engine
- IPs Found: 0
- Emails Found: 0
- People Found: 0
- Hosts Found: 0

Note: Baidu search results for microsoft.com were limited, resulting in no direct findings from this Chinese search engine source.

```
*****
[*] Target: microsoft.com
[*] Searching Baidu.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.

(fatima@kali)-[~]
```

5.3 Task 2: Reconnaissance with All Sources

Command Executed:

```
$ theHarvester -d microsoft.com -l 50 -b all
```

Results Summary:

- Total Subdomains Found: 60+
- Sample Subdomains:
 - eus.firstparty.entitlements-int.commerce.microsoft.com
 - eus.licensing.entitlements-int.commerce.microsoft.com
 - eus2-0.frontdoor.licensing.commerce.microsoft.com
 - invalidation-mr-prod-westus.cms.microsoft.com
 - iotcentral.microsoft.com
 - iotschool.microsoft.com

```
(fatima@kali)-[~]
$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [ASCII ART]
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
```

5.4 Findings & Impact

theHarvester successfully enumerated 60+ subdomains for microsoft.com from multiple public sources. Key findings include:

- Multiple subdomain naming patterns reveal internal organizational structure
- Licensing, commerce, and service subdomains expose business infrastructure
- Each subdomain represents a potential attack entry point

6. W2-PM5: NETWORK SCANNING WITH ZENMAP

6.1 Background

Zenmap is the official GUI (Graphical User Interface) version of Nmap, used for network discovery and security auditing. It can identify live hosts, map network topology, detect operating systems, and identify open ports. Zenmap works across Linux, Windows, Mac OS X, and BSD platforms.

6.2 Task Execution

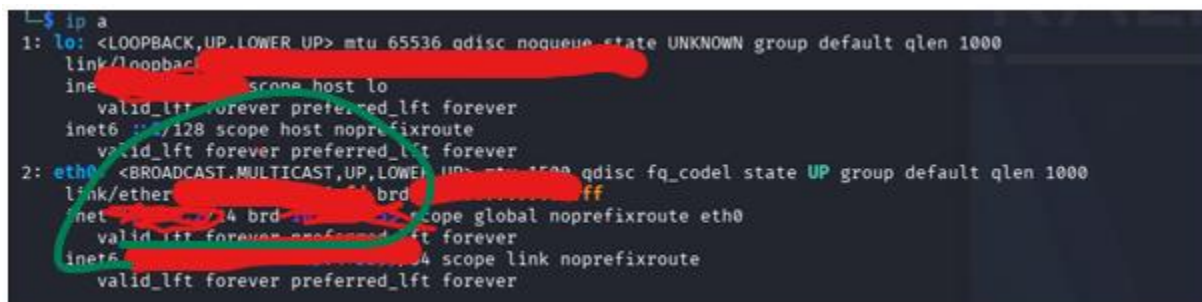
Task 1: Installation

Initially attempted to install Zenmap on Windows PC. Due to installation challenges, theHarvester was run in Kali Linux instead using the pre-installed Nmap/Zenmap tools.

Task 2: Local Network Configuration

Command Used:

\$ ip a



```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
    net 10.10.10.14/24 brd 10.10.10.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::208:10ff:fe00:0014 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Task 3-7: Network Scanning Results

Command Used in Zenmap:

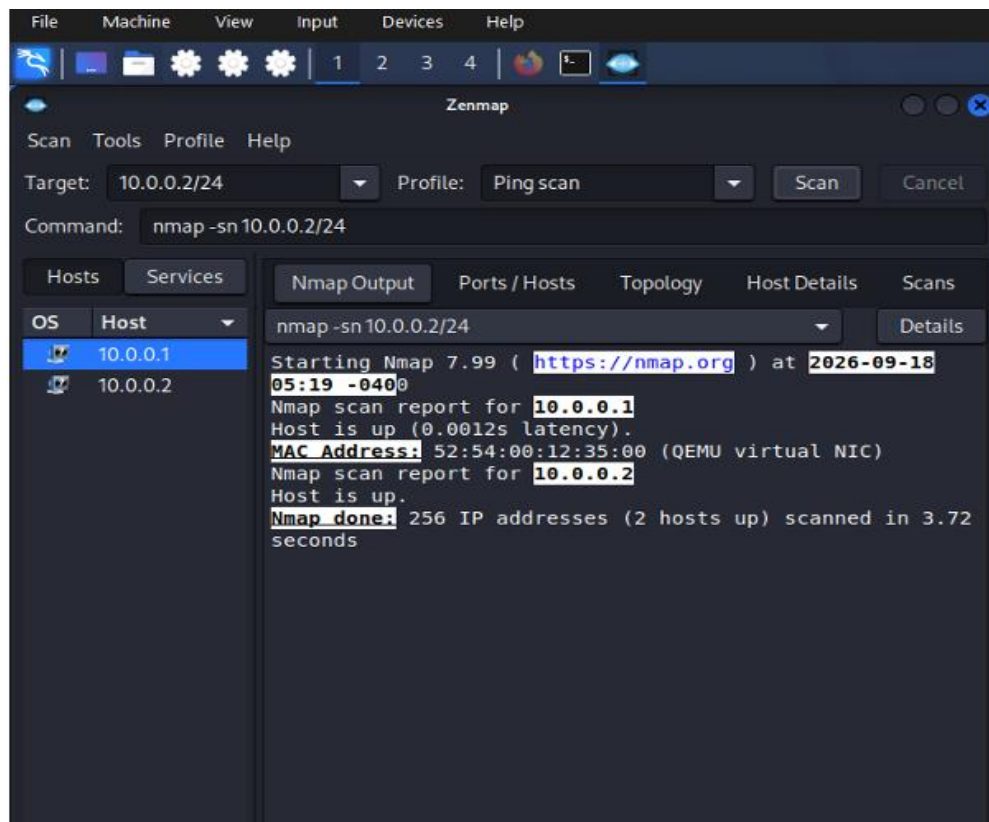
```
$ nmap -sn 10.0.0.2/24
```

Scan Results:

Metric	Result
Hosts Scanned	256 IP addresses
Scan Duration	3.72 seconds
Live Hosts Found	2 hosts

Live Hosts Discovered:

No.	IP Address	MAC Address
1	10.0.0.1	52:54:00:12:35:00
2	10.0.0.2	(Local machine)

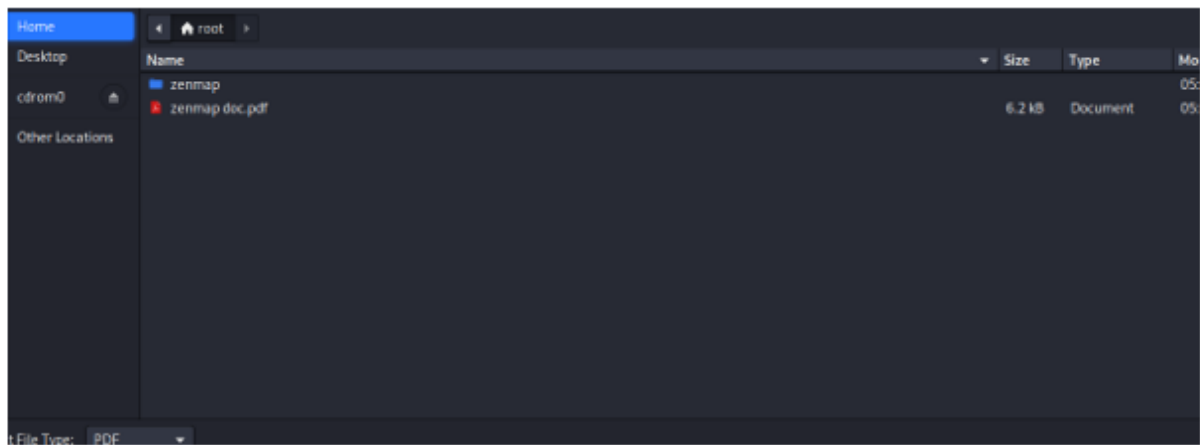


6.3 Topology Diagram

A network topology diagram was generated in Zenmap showing the relationships between discovered hosts. The topology was saved in PDF format for inclusion in this report.

6.4 Findings & Impact

- 2 live hosts were identified on the 10.0.0.0/24 subnet
- Full network mapping and topology generated
- MAC addresses obtained for ARP spoofing/VLAN hopping analysis
- Complete network baseline established for comparison with future scans



7. RISK ANALYSIS

Based on information collected during all three reconnaissance phases, the following risks were identified:

Finding	Impact	Risk
Exposed Security Cameras	Privacy breach, facility surveillance	HIGH
Open Directory Listings	Intellectual property theft	HIGH
Subdomain Enumeration	Attack surface mapping, infrastructure disclosure	MEDIUM
Live Host Discovery	Network mapping for further attacks	MEDIUM

8. RECOMMENDATIONS

8.1 For Organizations

1. **Audit Exposed Resources:** Regularly search your own domains using GHDB dorks to identify what you are leaking to Google and fix it before attackers find it.
2. **Remove Camera Access:** Ensure security cameras are behind firewalls and VPNs. Never expose them to the public internet.
3. **Configure Directory Listings:** Disable 'Index of' directory listing in web server configurations (Apache, Nginx).
4. **Robots.txt and Meta Tags:** Use robots.txt and noindex meta tags to prevent search engines from indexing sensitive content.
5. **DNS Hardening:** Minimize public DNS records. Use DNS filtering to only expose necessary subdomains.

8.2 For Network Defenders

1. **Regular Scanning:** Perform monthly network scans to identify unauthorized devices or changes.
2. **Network Segmentation:** Separate sensitive networks from public-facing infrastructure.
3. **Monitor DNS Queries:** Alert on suspicious DNS resolution patterns that might indicate reconnaissance.
4. **Implement NAC:** Use Network Access Control to ensure only authorized devices connect to the network.

9. CONCLUSION

During Week 2 of the Cybersecurity & Ethical Hacking internship, three comprehensive reconnaissance modules were completed successfully:

- ✓ GHDB Footprinting identified 20 exposed resources including security cameras and open directories
- ✓ theHarvester reconnaissance discovered 60+ subdomains for microsoft.com
- ✓ Zenmap network scanning mapped live hosts and created network topology

These exercises demonstrated that information gathering is the foundation of every successful attack. Even passive reconnaissance using only public sources can reveal extensive infrastructure details, security cameras, and organizational structure.

The most critical lesson is that organizations leak far more information than they realize. By regularly performing these same reconnaissance techniques on their own domains, defenders can discover what attackers see and reduce the attack surface before exploitation begins.

Security is a continuous process. Regular audits, prompt patching, proper configuration, and security awareness are essential to defend systems effectively against sophisticated attackers.

10. APPENDIX: TOOLS & COMMANDS REFERENCE

A. GHDB Workflow

1. Visit www.exploit-db.com
2. Click on GHDB section
3. Search for desired dorks (e.g., 'cam', 'index.of', filetype:pdf)
4. Copy dork query
5. Paste into google.com search bar
6. Review results and verify each link

B. theHarvester Commands

Single Source (Baidu):

```
$ theHarvester -d <domain> -l <limit> -b baidu
```

Multiple Sources:

```
$ theHarvester -d <domain> -l <limit> -b all
```

C. Zenmap Commands

Find Local IP:

```
$ ip a or $ ipconfig (Windows)
```

Ping Scan Subnet:

```
$ nmap -sn <subnet>/24
```

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